

Villanova University

Department of Mechanical Engineering

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**INTERIM REPORT 3. THERMAL FLUID DESIGN OF A DATA CENTER
RACK LIQUID COOLING SYSTEM**

Team 4:

Hooper, Henry

Tatum, Sam

Williams, Nick

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1. Introduction and Specifications

Data centers contain large servers that house and operate thousands of processing chips. Effective cooling systems are required to maintain safe operating temperatures. This Interim Report was focused on designing a rack-level cooling system for 18 servers by calculating flow impedance, necessary flow rates of coolant, coolant temperatures, and material usage. The cooling system needed to deliver cooling water from the pump in the coolant distribution unit (CDU) to the servers, then return the water to a buffer tank. The system's intake and exit manifolds before and after the server flow loops were designed to ensure equal coolant distribution to each rack while minimizing impedance.

The cooling system needed to distribute coolant to 18 servers that were stacked vertically in the rack. Thermal specifications for both the servers and the rack are shown below in Table 1.

Table 1. Server Rack Specifications

Parameter	Value
Server GPU total power per server	16 kW
Server low power component power per server	3.2 kW
Number of 1U Racks	18
Height of 1 Rack Unit	1.75 in
Standard Rack Width	19 in
Rack liquid loop total thermal load	288 kW
Rack air cooling total thermal load	57.6 kW
Data Center Air Temperature	20°C

The water needed to be equally divided into each server flow loop to ensure that the cooling specifications calculated in Project 2 could be maintained. Flow rate and temperature were the two main components in keeping the chip temperature as close to the ideal temperature as possible by allowing the cold plates to function as designed. To achieve constant and equal flow rates, an intake manifold was designed to divide the incoming coolant flow. The manifold was used to minimize the pressure drop throughout and prevent the liquid's velocity from changing, which would have resulted in varying flow rates supplied to each branch.

2. Overall System Cooling Requirements

Each server was designed to transfer 16 kW to the cooling fluid by way of eight cold plates, one for each chip on the server. In Project 2, the thermal-fluid needs of each server were calculated, and these were scaled to determine the overall needs for the entire cooling system. The individual server layout is shown below in Figure 1.

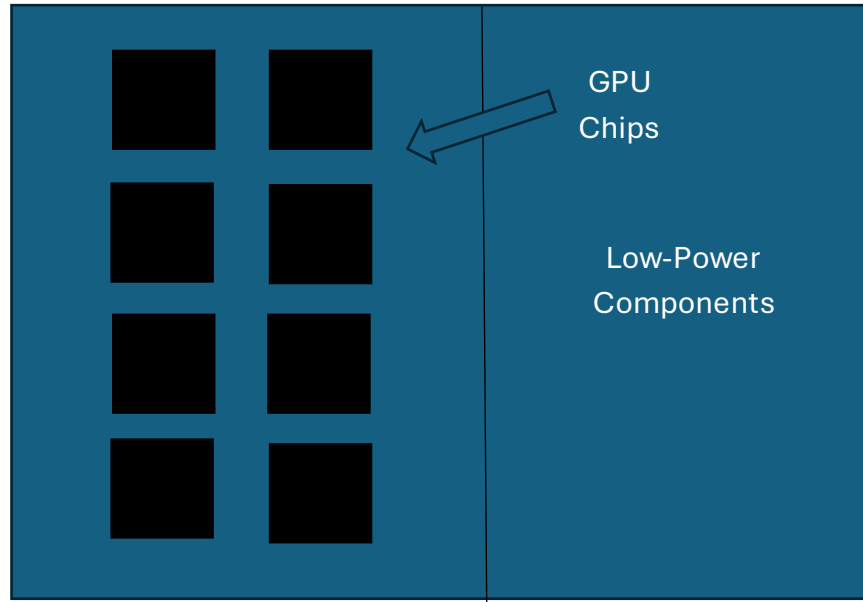


Figure 1. Server Rack Layout

The inlet water temperature was given at 20°C. The outlet temperature (T_o) was calculated using the energy balance shown below, where the total heat dissipated by the cold plate (q) was transferred to the water. The water was flowing at a mass flow rate ($\dot{m}$) of 0.02 kg/s, or 1.2 LPM, as designed in Project 2. The water had a specific heat (c_p) of 4182 J/kgK. ¹

$$T_o = T_i + \frac{q}{\dot{m} * c_p} \quad (1)$$

The outlet temperature was calculated to be 43.93°C. Project 2 also designed the flow loop for a single server, and that design required an 8-branch parallel flow to deliver 20°C water to each cold plate. Since all the cold plates were fed individually, the server's overall outlet temperature was 43.93°C.

Table 2. Flow System Thermal-Fluid Specifications

Parameter	Value
Flow Rate (One Server)	9.6 LPM
Flow Rate (Total Server Rack)	172.8 LPM
CDU Delivery Water Temp	20°C
Facility Temp to CDU	43.93°C
Air Temperature Leaving Rack	25.83°C

All of the system's water exiting the return manifold was at the same temperature as that observed at the hot water outlet from a single server. The water heated by the GPU through the Interim Report 2 cold plate design was calculated to exit at 43.9°C. Since all 18 servers' hot

water outlets were connected to the same exit manifold, all water in the manifold reached an equilibrium of 43.9°C.

The water temperature requirements at all points in the server rack and CDU were calculated, assuming that the fluid would not transfer any heat through the intermediate piping between components. No specification was provided on how the room was cooled. The air temperature exiting the RDHX and entering the room was calculated to be higher than the ambient air temperature (about 20°C). Buildup of hot air circulating through the server rack room would cause a gradual temperature increase over time. If no cooling systems (AC or vents) were implemented in the room, it would no longer be valid to assume that the calculated temperature requirement at the exit of the pump and the entrance of the inlet manifold were the same, due to ambient convection with the piping and subsequent conduction into the pipe. The room's air temperature would be hot enough to heat the pipes externally and increase the water temperature before it even reached the GPU cooling plates.

3. Design of Rack Cooling System

The first component designed was the intake and exit manifolds. This design was to ensure even distribution of flow into the server branches, allowing them to receive the optimal flow rate as specified in Project 2. To accomplish this, the manifolds needed to have a pressure drop that was much greater than that of any individual branch.

The staggering ratio (SR) was calculated in Equation 2 by dividing the branch spacing (L_B) by the branch diameter (D_B).

$$SR = \frac{L_B}{D_B} \quad (2)$$

Research showed that an optimal staggering ratio was 1.75.² The server racks were 1.75 inches tall, so the branches were staggered 1.75 inches apart to be level with the servers. Using the 1.75 ratio, the optimal branch diameter was 1 inch, a slight decrease from the 1.5-inch diameter pipe specified in Project 2. Based on this design change, the pressure drop through the branch was recalculated using the major pressure loss equation shown below, as a function of the friction factor (f), the speed of the fluid (U_w), the density of the cooling fluid (ρ), and a ratio of the pipe length (L) to diameter (D).

$$\Delta P_M = f \left(\frac{L}{D} \right) \left(\frac{\rho U_w^2}{2} \right) \quad (3)$$

The friction factor was found using the Moody Chart [A-1], based on the calculated Reynolds number through the pipe. The pipe experienced 9.6 LPM of water through a 1-inch diameter, yielding a Reynolds number of 8583. This resulted in a friction factor of 0.32, and the adjusted total pressure loss through the branch was 0.1378 psi.

The next design point was the length of the piping from the manifolds to the server loops. The server loops were designed to be 16 inches wide, and the server rack was 19 inches wide, so

there were 1.5 inches remaining on both sides. This was the determined length of the inlet and exit pipes to the manifolds, to minimize the frictional losses.

The manifold design was made considering the area ratio between the cross-sectional area of the branch and the cross-sectional area of the manifold header. Research showed that an area ratio of 0.14³ was optimal for ensuring that the branch pressure was much lower than the manifold pressure, ensuring equal division in the flow. The area ratio (AR) was computed using the ratios of the branch diameter (d_B) and the manifold diameter (d_M).

$$AR = \frac{d_B^2}{d_M^2} \quad (4)$$

Equation 4 was rearranged to solve for the branch diameter, using the previously specified branch diameter of 1 inch and an area ratio of 0.14. The optimal manifold diameter was solved to be 2.67 inches. The closest standardized pipe size is 2.5 inches, so this value was deemed optimal for the manifold diameter based on the area ratios and ease of manufacturing.

The inlet manifold height was designed to deliver water from the pump hose to the top server. Since there were 18 servers, each 1.75 inches tall, the total height of the inlet manifold was 31.5 inches. The height of the exit manifold was a similar design, but instead used the distance from the top of the server to the hose leading to the buffer tank. The height once again included the height of all 18 servers and totaled 31.5 inches.

Table 3. System Fittings Design Specifications

Parameter	Value
Dimensions (height, cross-sectional area) of the inlet manifold (in,in ²)	31.5, 4.908
Diameter of the exit ports from the manifold to the inlet hoses (in.)	1.0
Diameter and length of the inlet hoses from the manifold to the server inlet (in.)	1.5, 1.0
Dimensions (height, cross-sectional area) of the exit manifold (in,in ²)	31.5, 4.908
Diameter of the inlet ports to the exit manifold from return hoses (in.)	1.0
Diameter and length of the return hoses from server to manifold inlet (in)	1.0, 1.5
Diameter and length of the hose from the pump to the inlet manifold (in.)	2,12
Diameter and length of the hose from the exit manifold to the tank and to the RDHX (in.)	2,12

The hose diameters and lengths were based on research on the CDU size, to determine how far the water needed to be transported, as well as industry-standard hose diameters for high-volume water flow.⁴ The 2 in diameter hoses, which transported water at a flow rate of 172.8 LPM, yielded a water velocity of 1.42 m/s. The lengths were set at 12 inches from the pump to the manifold, and 12 inches from the exit manifold to the buffer tank. The final designed rack system is shown below in Figure 2.

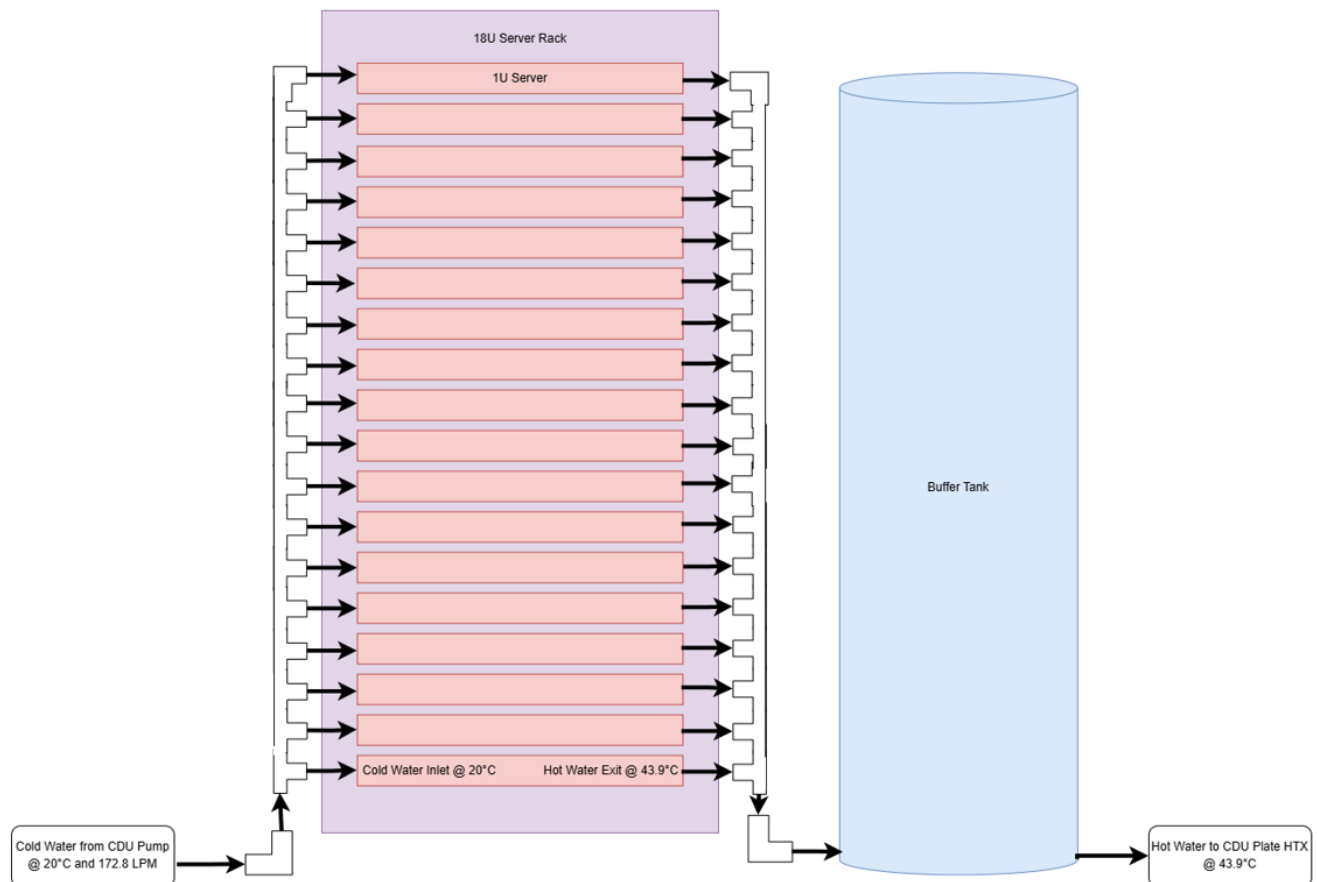


Figure 2. 18U Server Rack P&ID Diagram

The diagram for a single server flow loop is shown below in Figure 3. This was the flow loop designed in Project 2, with the only difference being the changed incoming branch diameter to fit with the manifold design. Additionally, the bill of materials for the system is shown in Figure 4, with possible purchasing sites and pricing for each component.

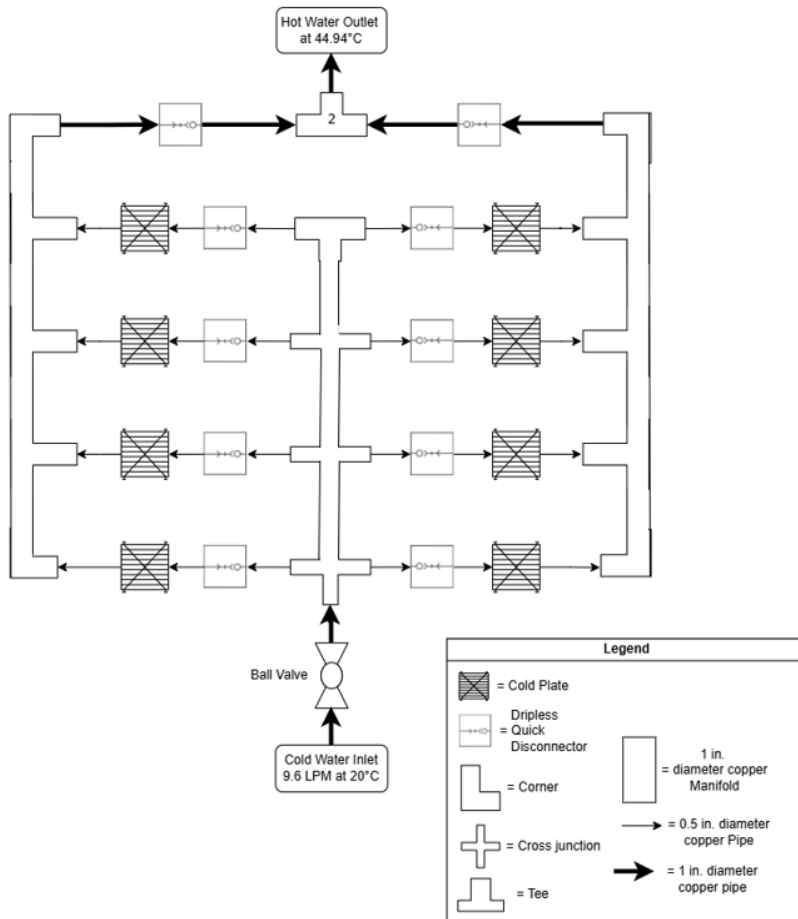


Figure 3. Individual Server Flow Loop

Fittings	Material	Diameter (in)	Length (in)	Quantity (#)	Price per unit (\$)	Part Number	Site
Elbow (90°)	Copper	2	5	2	82.73	PFXPP9K	PROFLO
Tee	Copper	1.5	5	18	76.99	PFXPPTJ	PROFLO
Ball Valve	Stainless Steel	1.5	7.59	18	480.00	SS-67TF24	Swagelok
Dripless Quick Disconnect Set	Stainless Steel	0.5	2	180	1812.00	SS-QDF8 / SS-QDC8	Swagelok
Manifolds							
Center Server Flow Loop Manifold	Copper	1.5	14	1	Quote required	MANI_1	Designed
Perimeter Server Flow Loop Manifold	Copper	1.5	16	2	Quote required	MANI_2	Designed
Inlet/Outlet Server Rack Manifold	Copper	2.5	31.5	2	Quote required	MANI_3	Designed
Components							
Cold Plate	Copper	N/A	2	144	Quote required	COLDPLATE_1	Designed
Piping							
Copper Tube	Copper	0.5	120	5	43.17	5175K123	McMaster
Copper Tube	Copper	1	120	3	102.92	5175K126	McMaster
Hosing	Flexible PVC	2	24	1	35.79	ZUSA-HT-3037	Grainger

Figure 4. 18U Server Rack Bill of Materials

4. Prediction of Rack Flow Impedance

The total impedance, or pressure drop, through the rack system was calculated as the summation of all the major head losses due to friction (ΔP_M), the minor losses (ΔP_m), and the gravitational losses (ΔP_g). The major losses were calculated as a result of the friction factor (f), the speed of the fluid (U_W), the density of the cooling fluid (ρ), and a ratio of the pipe length (L) to diameter (D). The velocity through each portion of the system relied on the flow rate through that component as well as the cross-sectional area.

$$\Delta P_M = f \left(\frac{L}{D} \right) \left(\frac{\rho U_W^2}{2} \right) \quad (5)$$

The minor loss calculations were due to the loss coefficients found for each component, as well as areas of expansion or contraction. The minor loss equations also accounted for the fluid density and velocity. Equation 6 calculated the minor head loss due to the components in the flow loop.

$$\Delta P_m = K \left(\frac{\rho U_W^2}{2} \right) \quad (6)$$

The gravitational losses were due strictly to the change in height of the fluid (Δz), the density, and the force of gravity (g). The only non-negligible height change in the system (not including the CDU) was due to the flow traveling through the manifolds. Equation 7 was used to calculate gravitational losses (ΔP_g).

$$\Delta P_g = \rho g * \Delta z \quad (7)$$

The total pressure loss accounted for several friction factors and loss coefficients, presented in Table 4.

Table 4. Applicable Friction Factora and Loss Coefficients

Component	Friction Value
Manifold Branch	K = 1.26
Branch Tee	K = 1.26
Vertical Manifold (2.5 in)	f = 0.023-0.039
Elbow (90°)	K = 0.63
Ball Valve	K = 0.05
Dripless Quick Disconnect	K = 2.5
Straight Copper Pipe (0.5 in)	f = 0.05
Straight Copper Pipe (1.5 in, 4.8 LPM)	f = 0.042
Straight Copper Pipe (1.5 in, 9.6 LPM)	f = 0.038
Rubber Hose	f = 0.02

The pressure drop across the manifold was due to the change in height, or hydrostatic pressure, and the friction losses. The hydrostatic pressure was a linear increase based on position through

the manifold, but the rate of frictional losses decreased as water in the manifold was directed into the branches. As seen above in Equation 12, the friction losses were related to the velocity, and since the flow rate decreased as water was diverted, but the size of the pipe did not, the velocity decreased. The change in pressure versus vertical position in the manifold is shown below in Figure 5.

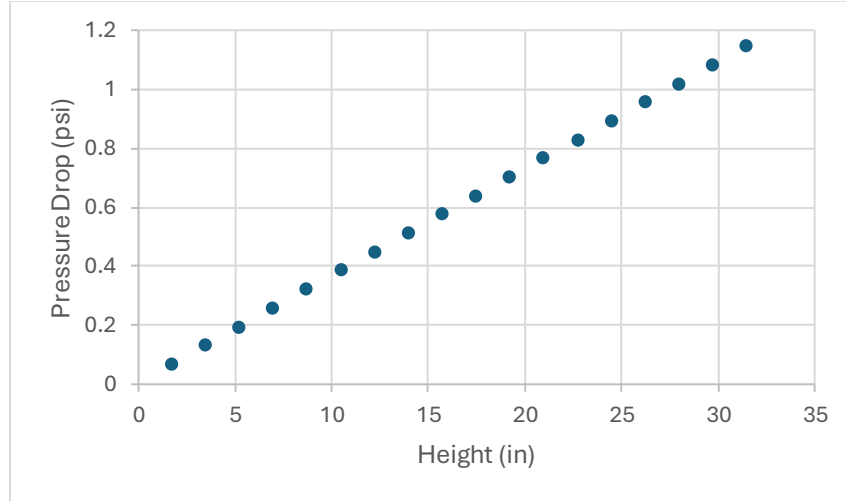


Figure 5. Pressure Drop vs Height in the Manifold

The pressure drop from the entire system was split into four main categories: the hosing, the manifolds, the server loops, and the cold plates. The losses due to the hoses were mostly due to pipe friction and a 90° bend to enter the vertical manifold. Additionally, there was a loss due to expansion from the 2-inch diameter hose to the 2.5-inch diameter manifold. The K value for this expansion was calculated as a function of the cross-sectional areas of the hose (A_H) and the manifold (A_m). The resulting K value was 0.13.

$$K = \left(1 - \frac{A_H}{A_m}\right)^2 \quad (8)$$

The losses in the manifold were due to friction throughout the manifold, as calculated in Figure 4 above. However, the pressure drop the water experienced due to an increase in height would be regained in the exit manifolds, so the only cause for pressure drop was the pipe friction and the losses through the branch inlets, which had a K value of 1.26. There were 18 branches in total from each manifold.

In the branches, the water experienced pipe friction and a contraction from the manifold to the smaller branch pipe size. The K value for this contraction was calculated to be 0.63, using the same Equation 15. The cold plate pressure losses were calculated in Project 2 and included losses due to the manifolds within the server flow loop, as well as the pipe bends and dripless quick disconnects. The total pressure loss contributions from each of these component groups were presented in Table 5 and were calculated at the target flow rate of 172.8 LPM.

Table 5. System Impedance at 172.8 LPM

Component Group	Value (psi)
Hose	0.860
Manifolds	0.691
Branches	0.698
Cold Plates	3.625
Total Impedance	5.974

In addition to calculating the system impedance at the target flow rate, an impedance curve was calculated for future use in designing a CDU. This CDU included a pump to overcome the system impedance at the target flow rate. The system impedance was calculated from flow rates of 100 to 280 LPM, which changed the water velocity through the components and increased friction and component losses. The impedance curve for this server system is shown in Figure 6.

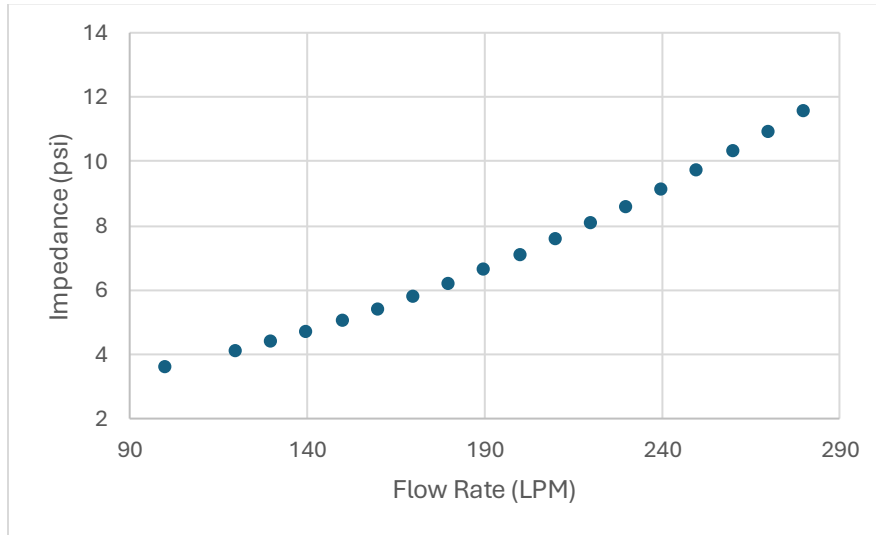


Figure 6. Impedance vs Flow Rate for Server Rack System

The relationship between pressure drop due to friction and flow rate was quadratic, where $\Delta P \propto Q^2$. This was consistent with the Ohm's Law equation for pressure losses due to pipe friction shown below, accounting for the friction factor (f) and the flow rate (Q).

$$\Delta P = f * Q^2 \quad (9)$$

5. Air Cooling System for Low Power Components

Each rack on the server unit contained the GPU chip array and an area for low-power components. The power dissipation in this section was less than that of the GPU array. The thermal load of the low-power components was specified as 20% of the total dissipation from the

main chip array. The low-powered nature of this section meant that air was a possible component cooling method. The parameters of the low-power components are shown in Table 6.

Table 6. Parameters of Low-Power Component Area

Parameter	Value
Width (w)	8 in
Length (L)	16 in
Height of Space Above Chip (h)	2 in
Thermal Load (q)	3200 W
Kinematic Viscosity (ν)	$1.57 \times 10^{-5} \text{ m}^2/\text{s}$
Thermal Conductivity of Air (k_a)	0.026 W/mK
Maximum Component Temperature	75°C
Incoming Air Temperature	25°C

The goal of this design was to keep the component temperature below 75°C. The one changing parameter was the flow rate and the velocity of the air passing over the components. First, the Reynolds number was calculated from a given air velocity (u_w), the kinematic viscosity for air (ν), and the length of the area (L), which was given as 16 inches. The velocity of the air was calculated as a function of the flow rate (V_a) and the cross-sectional area of the passage through which the air flowed.

$$Re = \frac{u_w * L}{\nu} \quad (10)$$

$$u_w = \frac{V_a}{w * h} \quad (11)$$

The Reynolds number was then used to calculate the Nusselt number for this specific situation. The Nusselt correlation also used Prandtl's number, which was given as 0.71⁵.

$$Nu = 0.037(Re^{\frac{4}{5}} - 871)(Pr^{\frac{1}{3}}) \quad (12)$$

With the Nusselt number calculated, the average convective heat transfer coefficient (h) could be calculated as a function of the Nusselt number, the conductivity of the air (k_a) and the length (L).

$$h = \frac{Nu * k_a}{L} \quad (13)$$

The component temperature (T_c) was then calculated using a convective heat transfer equation. The total rate of heat dissipation (q) was given, as was the air temperature (T_a), and the surface area of the low-powered components (A).

$$T_c = \frac{q}{h * A} + T_a \quad (14)$$

The component temperature was calculated for a range of air flow rates to determine if it was possible to cool the components below 75°C. The flow rates were graphed against the maximum temperature, as shown below in Figure 7.

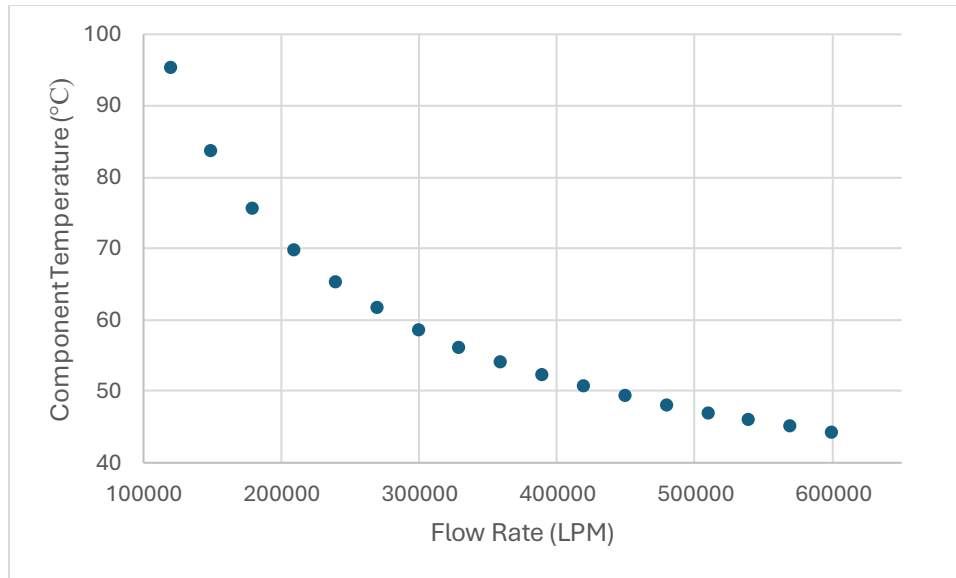


Figure 7. Component Temperature vs Air Flow Rate at 3200 W

The component temperature dropped below the required 75°C threshold at around 180,000 LPM. At this design point, the h value was around 765 W/mK, which was too high to be reached using air cooling. For an h value of 765 W/mK, the air had to be moving at 145 m/s, which is unreasonable and impossible for fans to achieve. If the total power output was decreased to 20% of a single GPU chip, then the air cooling became more reasonable, as shown below in Figure 8.

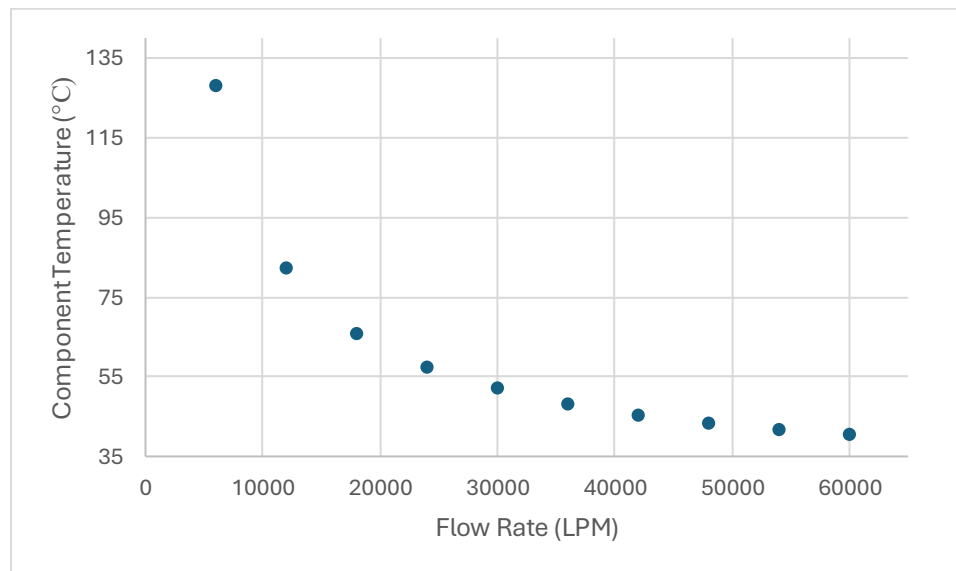


Figure 8. Component Temperature vs Air Flow Rate at 400 W

At this design point, the necessary h value was reduced to around 90 W/mK, which was much more reasonable using air cooling, only requiring an air velocity of around 12 m/s. This was still very high, but more reasonable than the value of 145 m/s required to dissipate 3200 W.

The result of this investigation showed that at 3200 W of power, the low-power components were not able to be cooled using air. At a lower power output, there was a possibility. To size a fan array to supply the necessary air flow, the pressure drop would also need to be found. Additionally, the layout of the fans would need to be addressed. Research showed that fans are usually arranged in parallel, and could be found in sizes of around 40mm, which would be able to fit in a single line across the component area to deliver air flow.⁶

The theoretical output temperature of the air (T_o) was calculated using an energy balance equation involving the specific heat of the air (c_p), the mass flow rate ($\dot{m}$), and the rate of heat dissipation (q).

$$T_o = \frac{q}{\dot{m} * c_p} + T_i \quad (15)$$

The output air temperature was calculated to be 25.86°C at 3200 W and 180,000 LPM, and 26.34°C at 400 W and 15,000 LPM.

6. Discussion and Conclusions

The goals of the server rack design were to provide equal flow distribution to every cold plate in every individual server. The designed cross-sectional area and branch ratio minimized the pressure losses through the branch as compared to the losses through the manifold. The pressure drop through the manifold was calculated to be 0.7 psi, whereas the pressure drop through each branch was only about 0.038 psi.

The impedance calculations were important to set up the next stage of design, the coolant distribution unit. Since the flow components and frictional forces create pressure drop and restriction, the chosen pump must be able to match the calculated impedance at the optimal flow rate. The calculated impedance of 5.87 psi at 172.8 LPM was deemed the target design point for the future, with an impedance curve also being calculated for varying flow rates to properly select a pump. Table 5 listed the calculated impedance for each component group in the system, and the numbers showed that the highest impedance resulted from the cold plates and server flow loops, which follows logically since the rest of the system was made of basic piping hosing and two manifolds. This total was also consistent with research into current server rack designs.

The cooling system for the low-power component area was deemed impossible to cool using air. The required heat transfer coefficient to cool the surface using 25°C air meant that the air velocity had to be 145 m/s. This was not possible for fans to produce, which was why the air cooling for 3.2 kW of heat dissipation was impossible. If the power dissipation had been lowered, then air becomes a viable solution. At 400 W, the required air velocity was only 12 m/s.

References

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Appendix

A. Moody Chart for Pipe Friction Factors

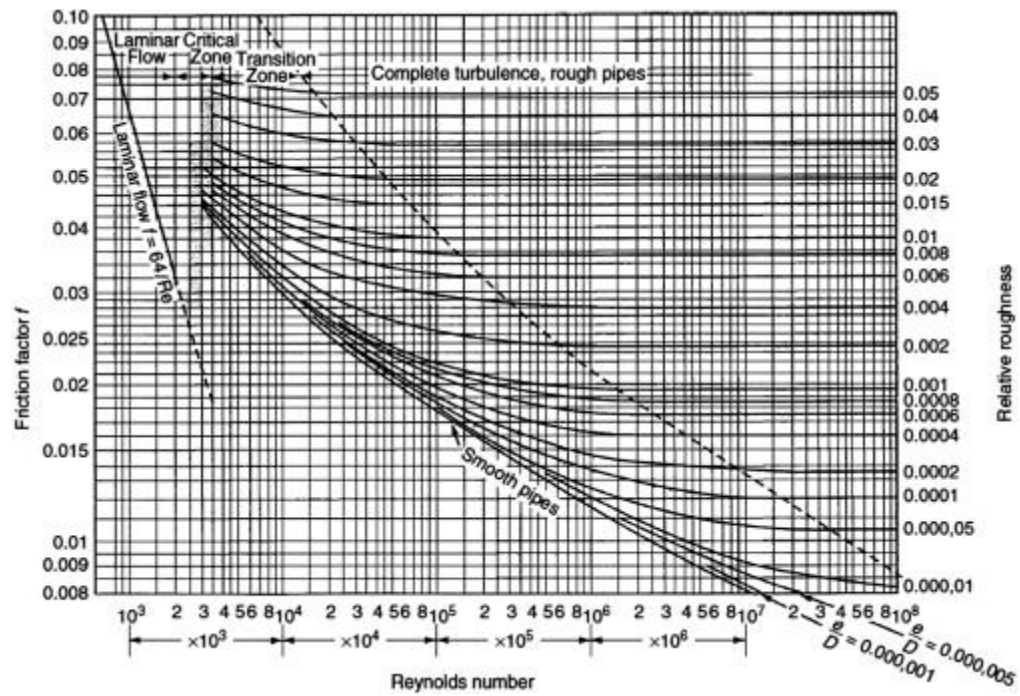


Figure A-1. Moody Chart